

PAPER_I_EXT_AUDIT — 03_split_falsification

Script 03 — Split-Graph Falsification of Theorem 1.1

Audit: PAPER_I_EXT_AUDIT · **Script:** 03_split_falsification.py

Environment: Python 3.14.4, scipy 1.17.1 (HiGHS LP), numpy · seed 20260710

Purpose

Directly test Theorem 1.1 — $|E(G)| - 2v^*(G) \leq n^2/6 + n$ — on split graphs, computing $v^*(G)$ **exactly** by solving the fractional triangle-packing LP on the whole graph (independent of the manuscript's two-phase construction). Also confirm the extremal family attains the leading constant.

Split graph model: clique K_p plus independent vertices with neighborhoods

$N(v) \subseteq K$. Triangles are clique triples and mixed triples $\{v, x, y\}$ with $x, y \in N(v)$

(no triangle uses two independent vertices, since I is independent).

Predeclared kill-switches

- **KS1** any split graph with $|E| - 2v^* > n^2/6 + n + 1e-6$ ■ **THEOREM FALSIFIED**.
- **KS3** extremal $K_p \vee K_{\lfloor 2p \rfloor}$ with $|E| - 2v^* \neq n^2/6 + n/6$ ■ tightness claim wrong.

Instances and results

- **(a)** structured enumeration $p=2 \dots 5$, one independent vertex per subset of K
(all neighborhood types): all satisfy the bound (ratios 0.25–0.30).
- **(b)** 600 random split graphs ($p=2 \dots 7$, up to 10 independent vertices, random neighborhoods): all satisfy the bound.
- **(c)** adversarial complete-split $N(v)=K$ with up to $3p$ independent vertices:
all satisfy the bound.
- **(d)** extremal family $K_p \vee K_{\lfloor 2p \rfloor}$ ($n=3p$), $p=2 \dots 7$: $v^* = C(p, 2)$ exactly, and

$|E| - 2v^* = n^2/6 + n/6$ exactly on every instance.

Quantity	Value
total split graphs tested (a,b,c)	622
theorem violations (KS1)	0
max ratio $(E - 2v^*) / (n^2/6 + n)$	0.815 (at $K_2 \vee K_{10}$)
extremal tightness failures (KS3)	0

Verdict

NOT FALSIFIED. No split graph violated $|E| - 2v_{\blacksquare}^* \leq n^2/6 + n$; the extremal family $K_p \vee K_{\blacksquare_{2p}}$ attains $n^2/6 + n/6$ exactly, matching the manuscript's §9 discussion (the leading constant $1/6$ is tight; the additive $+n$ is not optimized — observed peak $n/6$). **Limit:** a bounded computational search is supporting/typed evidence for a universal ($\forall$) claim, never a proof; the analytic proof (Scripts 01–02 + Gate H) carries the universal statement.